

# Don't Be So Negative: The Case for Negation-Free Syntax and Semantics\*

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## Abstract

Proof theorists in logic and mathematics have long been suspicious of the classical (Boolean) approach to negation. This suspicion began with work by Brouwer and Griss in the foundations of mathematics and extended to Heyting's proof theory for logic. Griss, in particular, inveighed against using negation in any form in proof theory. Setting aside questions about proof theory in mathematics and logic, what can we say about classical negation in a proof-theoretic semantics for natural language? Can we dispense with it? I argue that we can and, based on the evidence, we should. This suggests a corollary hypothesis: There is no genuine classical negation in the syntax of natural language. The words that we call "negations" ('no', 'not', etc.) are really just voiced indicators of something else altogether: abstract polarity markers – primitive  $\downarrow$  annotations on formulas that signal downward entailment environments and direct our inferential practice. Everything that linguists need to explain can be accomplished with those minimal resources and without classical negation.

## 0. Introduction

The classical view of negation (which here we understand to be the generally accepted account of negation after Boole) has largely been taken on board in the semantics of natural language, and by extension, in the syntax of generative linguistics. Classical negation is a view of negation as a bivalent, truth-functional complement operation on propositions. It flips truth value and leaves no third possibility. Formally, classical negation is characterized in the following way.

- $\neg P = T$  iff  $P = F$
- $\neg P = F$  iff  $P = T$

Typically, classical negation is also characterized by the following laws:

- Excluded Middle:  $P \vee \neg P$
- Non-Contradiction:  $\neg(P \wedge \neg P)$
- Double Negation:  $\neg\neg P \leftrightarrow P$

This approach to logic stands in contrast to the understanding of negation in intuitionistic mathematics and logic, where the law of excluded middle and the law of non-contradiction are typically rejected and where the negation of  $P$ , if it exists, is understood to not indicate falsity, but rather some other failure in the execution of the proof theory. Negation is more like an instruction of the form: Road Closed, go back in the other direction.

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Linguists have tended to endorse the classical view of negation in their semantic theories, and by extension in the syntax of natural language. There is also a belief that the syntax of natural language evinces classical negation in its morphemes and words. This position is perhaps driven more by our training in classical logic than by empirical evidence. There is no reason that semantics could not have been done in a proof-theoretic framework, and there is no reason that classical negation could therefore not have been abandoned or at least called into question.

There is certainly plenty of reason to think that negation in English and other languages is non-classical. If your boss tells you that she is “not *not* happy with your work,” she isn’t saying she is happy with your work. She is saying something in between. And while it is certainly a point of controversy, there may be all sorts of constructions in natural language that are neither definitively true nor false (a famous case would be Strawson’s (1950) contention that ‘The Present King of France is bald’ is, contra Russell (1905), neither true nor false).

One general motivation behind a proof-theoretic semantics is that you are interested in more than modeling entailment relations using set theory and possible worlds; you are interested in the formal mechanisms by which those entailments are derived – proved, if you will. Nothing in that idea immediately entails that classical negation must be rejected. It remains an option for proof theorists. Indeed, we will be looking at a proof-theoretic proposal in which negation does a lot of work. But when we begin from a proof-theoretic perspective, we are entitled to ask if classical negation is the best approach to the theory we are constructing. This is particularly true if we are offering a proof-theoretic semantics for natural language. If we take the semantics of natural language to be a theory of (idealized) human inferential competence, then questions arise about whether classical negation plays any helpful role in our enterprise.

The goal of this paper is to show that if we take a proof-theoretic approach to semantics, there isn’t much linguistic motivation for introducing classical negation; indeed, I am offering the hypothesis that there is no need for classical negation in either the semantics or syntax of natural language.

I will be working within a particular framework of proof-theoretic semantics or “natural logic” that was initially developed by medieval logicians and has been rehabilitated in contemporary logic in the form of polarity logics.<sup>1</sup> In the medieval version, the thought was that one could reduce all of the Aristotelean syllogistic rules to two basic rules, corresponding to two distinct environments, labeled *dictum de omni* and *dictum de nullo*. Today we would identify these environments as upward and downward entailing environments, each environment having its own substitution rule. We can illustrate the core idea in the following way: Any predicate *A* in a *dictum de omni* (upward entailing) environment can be replaced with a predicate *A'* that denotes a superset of *A*. Contrarily, in a *dictum de nullo* environment, any predicate *A'* can be replaced by a predicate *A* that is a subset of *A'*. Of course, from a proof-theoretic perspective, we don’t want to talk about sets, so we would express the idea by saying that given  $A \rightarrow A'$ , We can substitute *A* for *A'* in a *dictum de omni* (upward entailing) environment and *A'* for *A* in a *dictum de nullo* (downward entailing) environment. (We take the fat arrow in  $A \rightarrow A'$  to indicate some natural language version of “all *As* are *A's*,” “the *As* are *A's*,” “if it’s an *A* it’s an *A'*,” etc.).

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<sup>1</sup> See Sánchez Valencia, (1994) and Moss (2015) for surveys. See also Ludlow and Živanović (2022; ch. 2-3).

Step one in understanding this thesis is to think of the upward and downward entailing environments being annotated on predicates within natural language.<sup>2</sup> The second part – the central thesis of this paper – is that so-called negation is just one form of polarity marking in natural language. Or more accurately, what we call “negation” is just an element that explicitly introduces a downward polarity marker into the syntax of the language. So, for example, “Socrates is not a dog” gets rendered as “Socrates is ↓dog,” or if you prefer [↓dog (Socrates)]. This signals that ‘dog’ is in a downward entailing environment, nothing more. Thus, we can infer that Socrates is not a collie, a border collie, etc. Surprisingly, we will find that this is all we need. We don’t need classical negation. Nor do we want it.

I begin investigating this hypothesis by using it to challenge a key idea in my collaborative work with Sašo Živanović. In Ludlow and Živanović (2022), we offered a proof-theoretic account of natural language in which classical negation played an important role. Indeed, quantificational operators were simply devices to introduce connectives and negations. Negation was thus a foundational element of our proof theory. But that now looks like a mistake. My first step here will be to show that the use of negation in that theory can be replaced with polarity markers and nothing more. What we call negation is just one device among many signaling the presence of polarity markers.

In this respect, the proposal being offered stands in a robust tradition in linguistics – a tradition driven by the idea that what you think you see in the syntax is not necessarily what is there in the semantics. Linguists have long understood that the morphological elements in our language may be signaling the presence of something you didn’t expect – what Ana Szabolcsi, in a survey of such cases, has characterized as “disembodied” semantic elements.<sup>3</sup> They are disembodied in the sense that they are not embodied as they appear to be in the syntax. A better expression might be “difembodied,” because they are embodied in the semantics differently than they appear to be in the syntax. From the proof-theoretic perspective, we would say that they are not embodied in the proof theory as they appear to be embodied in the surface syntax – or at least not how we were taught that they were embodied in the surface syntax.<sup>4</sup>

Many of the cases surveyed by Szabolcsi involve negation. Ladusaw (1992), for example, argued that negative concord in many languages is just indicating a directional entailing environment. Chierchia (2006, 2013) took an alternative route to negative concord, suggesting that some negative concord items signal the presence of an “exhaustifier” operation. Zeijlstra’s (2022) work on abstract negation markers also follows in this tradition. Some things look like negations but are not, but classical negation still remains as a residual part of these proposals. I am going to make the case that these approaches made the same fundamental mistake as did Ludlow and

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<sup>2</sup> There are alternative ways to execute this idea, including the proposal that a predicate is in the scope of an odd or even number of negative polarity markers. To keep matters simple, I am going to assume that the polarity of a predicate is ultimately in some way attached to or associated with the predicate itself. This might involve a kind of “polarity phrase” or PolP, in the sense of Zeijlstra (2022). This would in turn involve a kind of “feature checking” familiar in contemporary generative linguistics. For the moment, we set aside these details.

<sup>3</sup> Szabolcsi (2017) collected examples in which what you think you see in the syntax is not what you get in the semantics: [https://annaszabolcsi.com/wp-content/uploads/2020/07/szabolcsi\\_disembodied\\_handout.pdf](https://annaszabolcsi.com/wp-content/uploads/2020/07/szabolcsi_disembodied_handout.pdf)

<sup>4</sup> What I am gesturing at here is the idea that the proof theory runs directly off of “surface syntax,” and that the elements being proposed here (polarity markers, not classical negations) are part of the syntax of natural language.

Živanović: They failed to see that classical negation was not necessary in these accounts. At the end of the day, they did not need to introduce an abstract (“disembodied”) version of abstract negation.

The idea of explicit polarity marking is broadly in the tradition of Dowty (1994) and its formal execution in Moss (2012). Dowty proposed an important role for polarity markers, offering the idea that negative concord elements are fundamentally just negative polarity markers that, in turn, function as bookkeeping elements in the proof theory of natural logic. But as in the cases mentioned above, Dowty still held on to classical negation, even when he was tagging it with a downward polarity marker. As should be clear by now, I am going to argue that Dowty just failed to make an important last step. You have the polarity markers. That is all you need. Forget the classical negations.

Other linguistic accounts of negation have taken aim at the way we think of it structurally (as a unitary operator, taking a proposition or predication as its argument). Although these views have taken aim at the structure of negation in natural language, they have not rejected classical negation per se. Examples include the negative copula analysis of negation in Horn’s (1989) *Natural History of Negation*, Schein’s (2016) adverbial account of negation in *Noughty Bits*, and Zeijlstra’s (2022) account of abstract negation. I will make the case that these accounts are, at a minimum, consistent with the idea that classical negation can be dispensed with. Each account can be recast using abstract polarity markers rather than classical negation.

We will also look at work by Szabolcsi and Zwart (1993) in which they appeal to an explanation of a linguistic phenomenon (“weak islands”) that seems to rely on classical negation stealthily entering in the semantics, if not in the syntax, in the form of complementation. Again, we will see that a more modest proposal was possible – one that recasts complementation in terms of polarity reversal rather than classical negation.

This account of negation is, of course, fraught with philosophical consequences, but I have moved those thoughts to a separate philosophical appendix, since my concern in this paper is with the technical linguistic challenges that this approach generates. While that is my principal concern, this is not to say that my survey of accounts of negation is intended to be exhaustive. The examples are rather offered here as illustrations of how one can approach analyses that lean on classical negation and show how those analyses work just as well, if not better, once negation is expunged. I begin, however, with the concerns about classical negation that were originally voiced by the intuitionist mathematicians.

## **1. A Tradition of Suspicion: Brouwer, Griss, and the Foundations of Mathematics**

The idea that negation is a problematic logical primitive is not new. It has a distinguished history in the foundations of intuitionist mathematics. L.E.J. Brouwer, in his 1907 dissertation and his 1908 paper “The Unreliability of the Logical Principles,” mounted a fundamental challenge to classical negation. For Brouwer, the classical law of excluded middle (that every proposition is either true, or its negation is true) was not a logical law at all but an unwarranted generalization

from finite cases to infinite mathematical domains. Classical negation carries with it the assumption that ‘not-P’ has determinate positive content: to deny P is to assert something. But for Brouwer, in the domain of constructive mathematics, all we are licensed to assert when we assert ‘not-P’ is that an attempted proof of P would lead to absurdity. This is a far weaker, more qualified notion than classical (again, Boolean) negation.

Brouwer’s terminology was admittedly not stable here. Van Stigt (1998) catalogs some of the ways Brouwer described his stand-ins for the role of negation: “absurdity,” “constructed impossibility,” “impossibility,” “incompatibility,” “difference,” and “contradiction.”

One way to take the general idea (however described) is that in constructing proofs, trying to establish some conclusion, one, on occasion, hits a wall that appears to be unbreachable. That wall might come in many forms. It might involve a conclusion that strikes you as clearly wrong, or obviously absurd, or not (you believe) consistent with your project in some fundamental way. In practical terms, the issue of falsity need not arise. You just mark the end of your branch of reasoning as bad and then use that mark of badness as a strategy to back your way out of the situation you reasoned your way into.

In natural logic/proof theoretic semantics, this approach makes a fair bit of sense. When we reason logically, we seldom have access to some external notion of what is true, but we do reason ourselves into positions that seem to be wildly untenable, and if we are honest and reliable in our reasoning, we flip the polarity on the chain of steps that got us there.

Brouwer’s position may seem radical, but it is far from the most radical position. George Griss, a student of Brouwer, took the skepticism further. In a series of papers beginning with his 1946 “Negationless Intuitionistic Mathematics,” Griss argued that even intuitionistic negation (the bad outcome approach) was illegitimate. His objection was grounded in what he called an executability requirement: mental constructions, in the intuitionist framework, must correspond to mental acts that an agent actually performs. But reasoning from a hypothesis one takes to be defective (which is what negation, even in the intuitionistic sense, requires) is not a genuine mental act. Griss therefore proposed a mathematics developed entirely without negation, in which negative statements are replaced by positive assertions of difference or apartness.

Not surprisingly, the reception of Griss’s program was mixed. Heyting (1955), in his assessment of Griss’s work, acknowledged the philosophical seriousness of the executability requirement but concluded that negationless mathematics was too impoverished: there are mathematical results that simply cannot be stated without negation in any recognizable form.

I am not here to endorse the programs of Brouwer, Griss, and Heyting. The domain I am concerned with is natural language inference, not constructive mathematics, and the motivation for eliminating negation that I will offer is entirely different. It is empirical. What I do want to take from Brouwer and Griss is the thought that the foundational role assigned to negation in classical (and even intuitionistic) logic is an artifact of a particular theoretical choice, not an inevitable feature of idealized human inferential practice. This is particularly so in the semantics of natural language. We should ask, why is truth and its negation so foundational to our

semantics, given that humans have so little (if any) direct access to truth and its negation? Is our reasoning not driven by what we have strong convictions for and against?

## 2. Rethinking “Negation” in Linguistics

As I noted in the Introduction, the central thesis of this paper is the idea that what we call “negation” in the syntax and semantics of natural language is better understood as a downward polarity marker, as in Moss (2012), which we will indicate via the downward arrow:  $\downarrow$ . The thought is that what we take to be a classical Boolean negation is in fact a marker of a downward entailing environment, and at times an indicator of how aggressively the downward entailment should proceed. I believe this is reflected in elements of our linguistic practice.

Specifically, it is interesting to note that downward entailing inferential patterns are reflected in the ways that we probe people’s uses of negation, to see how serious they are. If a friend says, “I’m not hungry,” we may offer a series of downward entailing probes to see how far they are committed to this entailment pattern: “How about Chinese food? Not even Peking Duck? Not even Peking Duck at Bakkan?”

What this suggests is a picture where the  $\downarrow$  marker, when introduced by surface “negation” or some other means (e.g. via negative concord and negative polarity items), doesn’t mean that the downward entailment is taken to exhaustion, but to a contextually determined limit.

This leads to a serious question about whether these inference patterns can, in principle, ever be taken to exhaustion. The medieval names for the polarity environments suggest that maybe you can. *Dictum de Omni* (‘omni’ meaning “all” or “everything”) suggests that if you keep entailing in an upwards direction, you ultimately end up talking about something satisfying a general predicate like ‘thing’. How do you go on from there? *Dictum de Nullo* (‘nullo’ meaning nothing”) suggests that you end up carving down the subsets until you eventually arrive at the empty set. Or at least you end up talking about individuals instead of kinds. Thus, in the de nullo case, I can say ‘Socrates is not a dog’, and run a chain of inferences to ‘Socrates is not a collie’, ‘Socrates is not a Scotch Collie’, to ‘Socrates is not Lassie’. In the upwards case we can move from ‘Lassie is a Scotch Collie’ to ‘Lassie is a Collie,’ to ‘Lassie is a dog’, to ‘Lassie is a canine’ all the way up to ‘Lassie is a thing’. Moving downward and upward it initially appears we ultimately terminate the inferences at some limit.

However, the situation is not so simple, for it assumes that these properties and individuals are not “open-textured” (in the language of Fredric Waisman) or “underdetermined” (an expression I used in Ludlow (2012)). As we begin to approach these boundaries, we begin to confront serious questions. On the upper end, we encounter questions about the extension or range of ‘thing’, and ‘everything’, and we will find that we will face choices when we begin to precisify what we mean by these expressions. Some of those precisifications will suggest a progressively broader understanding of what a thing is. And it must be noted that there are serious questions about whether you actually can quantify over everything (for discussion, see Shapiro and Wright (2006)). Take the set of everything and generate the power set from that. Did you not just create a larger set of “things” than you started with? On the downward trajectory, we can infer that

Socrates is not a Lassie stage, and from there, there is no end to how far down the inference might run (nanostages etc).

Fortunately, only philosophers and semanticists imagine that the entailment is to be taken to exhaustion. We carry inferences as far as practical and useful. Viewing matters through this lens, many familiar linguistic constructions and phenomena receive new and, in my view, more naturalistic interpretations. Let's review a few.

**Cancellability:** If negation were a primitive truth-conditional operator, it should not be cancellable without contradiction. But downward entailment can be canceled: "I'm not hungry – well, not for a full meal anyway." "I don't eat meat – except fish." "I never drink – except for beer and wine and sometimes tequila on special occasions." These "cancellations" aren't really cancellations; they are just telling you where you can stop with the downward entailments. The standard treatment of these constructions with classical negation is that we are walking back what we said, but this isn't right. We are just setting the limits on the downward entailments we are committed to.

Here is one more example: If I say, "I can't eat any more" and then order dessert, there's no logical contradiction; people understand exactly what is going on. But if negation were a classical  $\neg$ , this should be a flat-out contradiction. The ease with which we process this expressed desire for dessert reveals that what was asserted was a  $\downarrow$  marker with limited downward reach, not a universal truth-functional negation. Once again, the question is, did you just revoke what you said, or are you setting limits on how far the downward entailment should go. I am suggesting that the latter analysis is correct.

**Hedging patterns:** In "I don't really like jazz," the 'really' is doing work that wouldn't be sensible if negation were classical. It's signaling that the  $\downarrow$  propagation is not exhaustive; there may be some jazz that passes the filter. Classical  $\neg$  doesn't admit of degrees, but  $\downarrow$  markers do, depending on how far down the inference chain they propagate.

**Rhetorical reinforcement:** If negation were already fully encoded as classical  $\neg$ , reinforcement would be redundant: "I don't like jazz, not at all, not ever, not in any form." Each addition ('not at all', 'not ever', 'not in any form') is pushing the  $\downarrow$  reasoning further down the inferential chain, toward low-level downward closure. The reinforcement is not redundant; it's doing real inferential work, extending the reach of the  $\downarrow$  marker. Classical  $\neg$  would be incoherent in combination with such reinforcement because it already commits to the full complement.

**Cross-linguistic NC patterns:** In negative concord (NC) languages, the number of n-words that appear in a sentence often tracks the depth of the downward inferential pattern; the more n-words, the more strongly and explicitly the downward inference proceeds. This is what you would expect: languages differ in how explicitly they voice the  $\downarrow$  propagation through the inference structure, and NC languages voice it more completely than languages like standard English do. As we will see in a bit, sometimes NC markers come with rather specific information on the optimal depth and width of downward inference.

**Hedging under questioning:** “A: Did you enjoy the concert? B: I wouldn’t say I enjoyed it.” The indirect form reveals that the speaker is asserting ↓ without yet committing to aggressive downward inference; they’re leaving open whether some parts were enjoyable or whether some degree of enjoyment applies.<sup>5</sup> Compare: ‘I didn’t enjoy it’, in which a stronger downward inference seems to be invited. Or consider: ‘I didn’t enjoy it at all’, which suggests deeper downward inference. ‘I hated every minute of it’ runs deeper still. Each step represents deeper ↓ propagation.

**NPI’s as depth guides:** As we will see shortly, Ladusaw (1992) and Dowty (1994) both offered the idea that NPIs like ‘ever’, ‘at all’, ‘any’, ‘in the least’ are only licensed in ↓ contexts. Dowty suggested, correctly in my view, that NPIs provide bookkeeping annotations for natural logic – they tell us when we can begin a process of downward inference. One thought we might entertain is the idea that polarity markers can not only tell you that you are in a downward entailing environment, but they can also signal how far down the inferential path you should go. As just noted, in ‘I don’t like jazz at all’, ‘at all’ signals a deeply propagated ↓ pattern of inference; you are unlikely to find a kind of jazz that I like. ‘I don’t particularly like jazz’ is weaker and leaves open the possibility that there is a sub-type of jazz (e.g. cumbia jazz) that I like. NPIs can be signals as to how far down the ↓ inferential pattern you can go.

### The Jespersen Cycle as Evidence for Polarity Markers

The Jespersen Cycle is the cross-linguistically recurrent diachronic pattern in which a negative marker loses expressive force and is subsequently reinforced by a new element that takes over as the primary negative marker, and may itself begin to weaken as the cycle restarts. The canonical example is from French: classical Latin ‘non’ weakened to Old French ‘ne’, which was reinforced by expressions like ‘pas’ (step), ‘mie’ (crumb), ‘point’ (dot), ‘goutte’ (drop), ‘rien’ (thing), and ‘personne’ (person); *pas* subsequently became the primary negative marker in standard French and *ne* is now largely absent from colloquial speech, with *pas* itself beginning to show signs of weakening in some varieties.

The cycle has been widely documented across languages and language families (Germanic, Romance, Semitic, and others), and it is one of the most robust generalizations in historical linguistics. But its explanation has remained puzzling for accounts that treat negation as a classical logical primitive. If negation is a truth-functional complement operation (a Boolean flip that either applies or does not), then there is no obvious sense in which a negative marker can be strong or weak, deep or shallow. Classical  $\neg$  has no scalar dimension of its own. It cannot bleach in the way the Jespersen Cycle requires, and the theory has no resources to explain why reinforcing elements are always recruited from the bottom of scales rather than from anywhere else in the lexicon.

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<sup>5</sup> This construction may also lend itself to an interpretation of introducing metalinguistic negation (e.g. “I wouldn’t put it that way”). I set this interpretation aside for the moment, noting that it is surely also available.

The polarity marker framework offers a direct explanation of the cycle and its characteristic features.<sup>6</sup>

### **What the reinforcing elements have in common**

As noted above, the elements recruited as reinforcers in the Jespersen Cycle are typically expressions that suggest minimal quantities, minimal units, or ontological floors: ‘pas’ (a step – a small unit of forward motion), ‘mie’ (a crumb — a small particle of food), ‘point’ (a dot – a small spatial unit), ‘goutte’ (a drop – a small unit of liquid), ‘rien’ from Latin ‘rem’ (a thing – hypothetically the most general entity, the ontological floor), ‘personne’ (a person – the minimal unit of the human domain), ‘jamais’ from Latin ‘iam magis’ (ever – hypothetically the exhaustion of the temporal dimension). In English, the same pattern appears in negative polarity items: ‘a wink’, ‘a finger’, ‘an inch’, ‘a damn’, ‘a fig’, ‘a hoot’. What these expressions share is not membership in a semantic field or a grammatical category but a structural property: they all indicate a lower end of some ordering – a scale of quantities, a scale of sizes, an ontological hierarchy, or a temporal dimension. They are recruited as reinforcers precisely because they indicate a serious breadth and depth of inference. The speaker thus uses them to signal that the ↓ inference is being taken seriously.

This is what the polarity marker framework predicts. ↓ is always relative to an ordering, and its inferential depth depends on how explicitly a lower range of that ordering is signaled. A negative marker that simply annotates a predicate with ↓ makes no explicit commitment about how far the inference runs. In reality, the depth is contextually determined and may be shallow. A reinforcer that names the lower end of a scale makes the depth explicit and maximal – the inference is being taken to a substantial, and here is a candidate bottom to prove it. The reinforcer is a depth intensifier that converts shallow conventional ↓ into a deeper ↓ commitment.

### **What bleaching is**

The standard description of the Jespersen Cycle says the negative marker loses force or weakens over time. This description is accurate, but it does not explain what force consists in, or why the loss is directional and recurrent. On the polarity marker account, what bleaches is specifically the depth-indicating function of the reinforcing element. As the expression grammaticalizes into a routine negative marker, speakers stop actively computing the scale or ordering that the expression originally invoked. ‘Pas’ eventually ceases to invoke the lower scale end of steps. ‘Rien’ eventually ceases to invoke the ontological generality of things. The expression becomes a conventional signal that ↓ is active, without specifying how far it goes.

Bleaching is therefore not semantic erosion in some general sense. It is specifically the loss of the lower-end-marking signal – the loss of the explicit commitment to taking ↓ to some

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<sup>6</sup> To be clear, the Jespersen Cycle is offered as evidence for the genuine gradience of negation strength – that is, as evidence that ↓ depth is a scalar dimension along which negative markers can be located more or less explicitly – rather than merely as evidence for variability in the syntactic encoding of a binary operator. A binary truth-functional operator like classical ¬ has no scalar dimension and therefore cannot exhibit the continuous bleaching the cycle requires; bleaching is only explicable if what is varying is inferential depth, which is a gradient property of the polarity marker itself. I am grateful to Anna Szabolcsi (personal communication) for pressing for clarity on this point.

significant inferential depth. What remains after bleaching is bare ↓ direction without explicit depth content. The marker still annotates the predicate with inferential direction but no longer makes any claim about how far that direction runs.

This explains several features of the cycle that are otherwise puzzling. First, it explains why the cycle is directional. Bleaching runs from explicit depth marking toward shallow conventional ↓, never the reverse. You cannot acquire bottom-marking function through grammaticalization; grammaticalization always converts pragmatic and emphatic content into a bare directional signal, which is precisely what strips the lower-end-marking away. The cycle runs one way because ↓ depth can only shallow through conventionalization, not deepen. Deepening always requires recruiting a new lower-end-marking expression, which is what stage 2 of the cycle does.

Second, it explains why reinforcers are always recruited from the lower end of scales. A speaker who wants to signal that a ↓ inference is being taken seriously needs an expression that gestures towards the bottom – towards some notional minimal unit, towards the operative ontological floor, or towards the exhaustion of some temporal or spatial dimension. Expressions from the middle or top of scales would not serve this purpose. The bottom-recruiting pattern is not a lexical accident but a consequence of what depth marking requires.

Third, it explains why the cycle is recurrent and in principle endless. There is no final reinforcer that resists bleaching, because any expression, once it becomes a grammaticalized routine marker, stops signaling depth of inference and eventually just marks direction. The cycle restarts because bare ↓ without depth-marking is expressively weak, and speakers reach for a new depth-marking expression to restore the depth that was lost. This is not a quirk of French phonology or an accident of language contact. It is a direct consequence of the tension between what ↓ depth markers need to do and what grammaticalization always does to them.

### **Depth markers and their scales**

The reinforcing expressions recruited in the Jespersen Cycle are not all doing the same thing. There are at least two distinct kinds of depth-marking at work. Scalar minimizers like ‘pas’, ‘mie’, ‘point’, and ‘goutte’ mark a serious commitment to a lower end of a quantity scale – to a very small (perhaps contextually minimal) unit of some measurable domain. Their depth-marking force comes from pointing to a small unit. The scale has a natural metric and the minimizer names its contextually smallest unit. This isn’t to say that the scale can’t be later expanded to a greater depth via half-steps etc. It is just to say that in this context, your polarity marker establishes a downward entailing environment to a depth that is serious in the current context. Again, as noted earlier, it is never possible to bottom out completely.

Ontological breadth markers like ‘rien’ work differently. Latin ‘res’ (thing, matter, entity) is not a minimal unit on a quantity scale. It is the most general noun in the context, one that signals the broadest category of entity relevant to the context. ‘Rem’ marks the breadth of the ↓ inference not by pointing to a scalar minimum but by pointing to the ontological range – the suggestion is that nothing sneaks through the downward entailing net.

This distinction matters because it shows that bottom-marking can be achieved through two structurally different operations – pointing to the contextually smallest unit on a scale, or pointing to the contextually most general level of an ontological hierarchy – and that the Jespersen Cycle recruits both kinds. What they share is not a semantic type but a structural function: both mark the contextual limits of some inferential direction.<sup>7</sup>

The thesis that there is no classical negation in either the syntax or semantics of natural language, it is admittedly strong, and it would be a challenge to prove, given all of the world’s languages and all of the places where people have imagined classical negation to appear, so what we really have operating here is a thesis and a research program. The thesis is that classical negation is overkill for what we want to explain in syntax and semantics. But at the same time, it is

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<sup>7</sup> Szabolcsi (personal communication) notes that historical linguists have identified multiple cycle phenomena beyond the negation cycle — including a reflexive cycle, a pronoun/agreement cycle, and a definiteness cycle — and gestures at the desirability of a unified account. The observation is well taken and worth pursuing, even if a full unified account lies beyond the scope of this paper.

The structural shape of these cycles is strikingly similar to the negation cycle. In the reflexive cycle, emphatic self-intensifiers like English *himself* (originally expressions with strong agent-anchoring force) grammaticalize into obligatory reflexive markers, bleaching their emphatic content as they become grammatically routine. Speakers then recruit new intensifiers to restore the emphatic force the grammaticalized form can no longer provide, and the cycle repeats. In the definiteness cycle, demonstratives with strong proximity-pointing force grammaticalize into definite articles, losing their pointing specificity as they become obligatory, and may grammaticalize further into agreement morphology. The pronoun/agreement cycle has the same shape: strong pronouns with specific referential force grammaticalize into agreement clitics and eventually into agreement morphology, with progressive bleaching of referential specificity at each stage.

In each case the cycle has the same three-stage structure: an expression with genuine emphatic or specificity-marking force grammaticalizes into a conventional grammatical signal, loses the expressive content that made it emphatic, and is eventually replaced or reinforced by a new expression recruited to restore what was lost. This structural parallel to the negation cycle is real and worth noting.

However, it is not clear that the same mechanism drives all of these cycles. In the negation cycle, what bleaches is specifically inferential depth – the scalar bottom-pointing content that signals how far the ↓ inference runs. This is a property that the ↓ framework can formally characterize. The reflexive and definiteness cycles involve a different kind of expressive content: agent-anchoring specificity and proximity-pointing specificity respectively. These are referential rather than inferential properties, and it is not obvious that the ↓ framework extends to cover them in any direct way. Calling demonstratives or emphatic reflexives “depth markers” in the proof-theoretic sense would be misleading. One could flip this analysis and say that the polarity markers involve some sort of direct reference to (deep) scale positions, but is not obviously true, and would require subtle argumentation.

The more modest claim is this: the negation cycle is one instance of a broader grammaticalization pattern in which emphatic expressions with specific expressive content bleach into conventional grammatical signals, with the lost content triggering recruitment of new emphatic expressions. The ↓ framework explains the negation instance of this pattern through the specific mechanism of inferential depth shallowing. Whether analogous formal mechanisms can be identified for the referential cases – some notion of referential depth or pointing specificity that admits of the same kind of gradient bleaching – is a genuine open question. Szabolcsi’s observation points toward a broader research program connecting grammaticalization theory with formal semantics and proof-theoretic approaches, in which the negation cycle treated here might serve as a model case. That program is a project for future work.

underpowered in its ability to describe our inferential strategies. The research program is to show, on a case-by-case basis, why we don't need (or want) classical negation in our syntactic and semantic proposals and how we might expand our descriptive and explanatory range by viewing polarity marking as part of a proof-theoretic project. Let's look at some examples to illustrate how this research program might proceed.

### 3. Revisiting some Proposals in Linguistics

Linguists have long been suspicious of the idea that voiced negative elements such as 'not', 'no', 'nobody', and 'nothing' behave as classical negation operators in logic. The suspicions have come from several independent directions and converged on a picture very close to the one defended here. In each case, the critic identifies a real problem with the classical analysis but stops short of the full dissolution that the polarity marker framework provides.

There is no shortage of proposals in natural language semantics that assign a central role to negation. The question is, do we really need classical negation for these proposals to work, or could those proposals fare just as well with a much more austere, Brouwer-esque alternative to classical negation? As we are going to see, there are a number of proposals in the semantics of natural language that lean on negation, but as we will also see, the same proposals could have been made without classical negation. One of the clearest examples of this is my collaborative work with Saso Živanović in our (2022) book *Language, Form, and Logic*. In that work, classical negation is explicitly built into the foundations of our proof theory. Everything runs through negation – or more precisely, through the presence and absence of negation as the determinant of logical polarity, which in turn governs the inferential operations our Dynamic Deductive System (DDS) permits. Let's see how negation can be dispensed with in that theory.

#### 3.1 Ludlow and Živanović's Dissolution of Quantification via Negation

The idea in Ludlow and Živanović is that negative polarity environments can be identified, not by polarity marking, but by being in the scope of an odd number of negations. Similarly, positive polarity environments are identified with being in the scope of an even number of negations. Notoriously, the antecedent A of a conditional 'if A then B' is a downward entailing environment. But if the conditional is placed in disjunctive normal form, the conditional is rendered as  $\neg A \vee B$ . The thought then is that all dictum de nullo (downward entailing) environments in natural language reflect the presence of a (disembodied) classical negation in the logical form of natural language. Ludlow and Živanović offer a formal language  $L^{**}$  which incorporates these properties and then argue that  $L^{**}$  is effectively the logical form (LF) of natural language.

On the L&Z picture, so-called quantifier expressions are not logical operators themselves but features that signal the presence of negations and logical connectives, and it is the resulting negations that establish polarity. So, for example, 'every' and 'no' signal negation (and thus downward-entailing environments) in their restrictions. (In the case of 'every A is B' the restriction is the A position and the scope is the B position). 'No' introduces a negation in its

restriction (and thus a downward-entailing environment) but also in its scope as well. These negations are what generate the polarity signatures that drive inference. Negation, in this picture, is genuinely foundational; it is not a derived notion but the primitive from which polarity, and hence the entire proof theory, is constructed.

This makes L&Z the ideal starting point for an argument that negation is dispensable. If negation can be eliminated here – in a framework where its foundational role is maximally explicit – then the argument for elimination is strong. And I will argue that the tools for that elimination are already hiding in plain sight in the L&Z framework itself, waiting to be noticed.

There are several elements that will be relevant to the discussion that follows. The first is the L&Z analysis of quantificational structure. The headline idea of that proposal is that there are no overt quantificational structures, and that what appear as “quantifiers” in the syntax are just features that control the construction of the surrounding clauses and introduce negations as appropriate. What remains are free variables which undergo quantificational closure, developing an idea from Ludlow (2002). In Linguistics, this has long been a familiar idea in the case of indefinite descriptions.<sup>8</sup> Free variables in conjunctive settings are interpreted as existentially closed. What the L&Z picture adds is the idea that if the restriction is in the scope of negation (in their system), the free variable is interpreted as universally closed. So, for example, in the structure  $[\neg Ax \vee Bx]$ ,  $x$  will be closed under universal quantification.<sup>9</sup>

As impoverished as such a system might appear, L&Z show that once it is wedded to a system of plural quantification and cardinal predicates (number phrases) it can do the full work of generalized quantifier theory. Indeed, they extend the result to infinitary languages. (Readers seeking details are referred to the book.)

What I am proposing here is that we can dispense with the negations and get by with *polarity form* alone. All the relevant semantic information is constrained in that form. Specifically, one can say that determiners like ‘all’ and ‘some’ and even the morpheme ‘not’ are voiced elements that introduce polarity markers and some structural information. The resulting picture can be seen in the following chart, which illustrates how determiners appear in natural language, their analysis in first order logic, their analysis in the L\*\* logical form of Ludlow and Živanović, and finally, once negation is abandoned, in the polarity form I am proposing here.

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<sup>8</sup> See, for example, Diesing (1992), and also much work in discourse representation theory.

<sup>9</sup> This is in the tradition of Szabolcsi (2015), which also treats quantifier particles as playing a role in building interpretive structure (in the semantics, in Szabolcsi’s case). It may appear that there is some conflict in approaches here, because Szabolcsi notes that the same particles are used for conjunction and universal quantification, and disjunction and existential quantification. This appearance of conflict stems from where the explanatory structure is working. From a semantic point of view, universal quantification does work as a possibly infinite conjunction. But one way to achieve the same result through syntactic means is by utilizing the  $\neg Ax \vee Bx$  structure proposed by Ludlow (2002) and L&Z. Every object  $x$  must satisfy that structure if it is true.

**Table 1. Polarity decomposition of quantificational forms**

| Surface form           | First order logical form                                           | L** form               | Polarity form                      |
|------------------------|--------------------------------------------------------------------|------------------------|------------------------------------|
| <b>Some A is B</b>     | $\exists x(Ax \wedge Bx)$                                          | $Ax \wedge Bx$         | $Ax \wedge Bx$                     |
| <b>Every A is B</b>    | $\forall x(Ax \rightarrow Bx)$                                     | $\neg Ax \vee Bx$      | $\downarrow Ax \vee Bx$            |
| <b>No A is B</b>       | $\neg \exists x(Ax \wedge Bx) / \forall x(Ax \rightarrow \neg Bx)$ | $\neg Ax \vee \neg Bx$ | $\downarrow Ax \vee \downarrow Bx$ |
| <b>Some A is not B</b> | $\exists x(Ax \wedge \neg Bx)$                                     | $Ax \wedge \neg Bx$    | $Ax \wedge \downarrow Bx$          |
| <b>Not all A are B</b> | $\neg \forall x(Ax \rightarrow Bx) / \exists x(Ax \wedge \neg Bx)$ | $Ax \wedge \neg Bx$    | $Ax \wedge \downarrow Bx$          |

This raises the question of why this pattern of quantifiers? Why not others? On the semantic side, in generalized quantifier theory, we know the answer to this question: One defines the class of possible determiners in terms of determiners that are model-theoretically conservative. One way to think about conservativity is that for any determiner DET, [DET As are Bs] is true if and only if [DET As are As that are Bs] is true. Determiners like ‘all but the As’ are ruled out because they are not conservative – they introduce new elements in addition to the As. L&Z show that you can also give a syntactic/proof-theoretic version of conservativity, called *restrictedness*. It turns out that the structures satisfying conservativity are those in which the disjunctions have negative polarity in their restriction and the conjunctions have positive polarity in their restrictions. You may have noticed that all the determiners given above satisfy that definition of restrictedness. Positive polarity restrictions occur with  $\wedge$  and negative polarity restrictions occur with  $\vee$ . However, this is not a mere stipulation in the L&Z framework.

If you think of logical inference as involving a process of cutting and pasting material in an argument structured as a tree, there is a rule governing what can be cut and pasted where. That rule, called *p-scope* by Ludlow and Živanović, is under the control of polarity, and the simplified version of the proposal is that disjunctions and conjunctions in the tree serve as polarity gates. You have to have negative polarity to pass through a disjunction, and positive polarity to pass through a conjunction. Restrictedness turns out to be a byproduct of that rule. The structures satisfying restrictedness are those in which the disjunctions have negative polarity in their restriction and the conjunctions have positive polarity in their restrictions. A syntactic item A cannot be recopied aside B in  $[A(x) \vee B(x)]$  because A(x), having positive polarity, cannot pass through the disjunction. Similarly, in the structure  $[\neg Ax \wedge Bx]$ , A(x) now having negative polarity, cannot pass through the conjunct.

Once we see that quantifiers dissolve into polarity markers and junctions, the dissolution of morphologically explicit “negations” (as well as other so-called n-words) becomes the natural next step. As we have already seen, ‘No A is B’ is  $[\downarrow A(x) \vee \downarrow B(x)]$ . Predicate negation (‘A is not B’) is  $[A \text{ is } \downarrow B]$ . ‘Is not the case’ is  $[\downarrow A \text{ is } \downarrow B]$ . Negation, in all its guises (the negative

quantifier, the negative copula, the sentential operator) turns out to be a family of polarity configurations, not a single primitive.

This may seem counterintuitive.  $[A \text{ is } \downarrow B]$  Seems so weak. Don't we need to definitively assert that A is absolutely NOT a B in any way shape or form? But this sentiment is misplaced. What  $[A \text{ is } \downarrow B]$  is telling us is that if you try to follow the inferential pattern to exhaustion (appropriating a term from Gennaro Chierchia here), you will see that no matter how much you shrink B, the downward entailing marker remains. In the abstract, there is no subset of B that satisfies x in 'A is x'. However, as we saw earlier, when we introduce polarity markers, we are seldom if ever committing ourselves to following the inferential pattern to absolute exhaustion.

### **3.2 Zwart's taxonomy of NPIs**

As noted earlier, what we take to be negations are downward polarity markers, but they are merely one among the many downward polarity markers that we find in natural language. We have already seen how determiners can introduce such markers. And it is important to recognize that negative polarity items (NPIs) also introduce these markers. We have already discussed some cases of NPIs, and I suggested that they are often signaling a certain aggressive downward entailing inferential pattern. However, it is worth noting that there are known taxonomies of NPI's that initially seem to have more to do with structural conditions than to do with morphological clues as to the depth of propagation of a downward entailment.

Zwart (1998) offers a fascinating cross-linguistic taxonomy of negative polarity items (NPIs) that suggests that some of the NPIs are sensitive to particular model-theoretic environments, typically those that are set-theoretically anti-additive and/or anti-multiplicative. The result is a taxonomy of NPI "strength": weak, medium/strong, and superstrong NPIs. What is interesting is that this taxonomy appears to track structural conditions more than contextually determined depth of downward inference. Mileage may vary on the data here, and it has led to some debates, but my goal at the moment is not to defend the taxonomy; it is just to show how it maps on to polarity form.

**Table 2 NPI strength and polarity forms**

| Surface category                                      | Model theoretic characterization                                  | Polarity form                           | NPI strength                 | English NPIs                                         | Dutch NPIs                                        |
|-------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------|------------------------------|------------------------------------------------------|---------------------------------------------------|
| additive & multiplicative                             | $f(A \cup B) = f(A) \cup f(B)$ and $f(A \cap B) = f(A) \cap f(B)$ | $\exists[\uparrow \wedge \uparrow]$     | none                         | —                                                    | —                                                 |
| additive & anti-multiplicative                        | $f(A \cup B) = f(A) \cup f(B)$ and $f(A \cap B) = f(A) \cup f(B)$ | $\exists[\uparrow \wedge \downarrow]$   | <b>weak</b> (in scope)       | ever, yet                                            | ooit                                              |
| multiplicative                                        | $f(A \cap B) = f(A) \cap f(B)$                                    | $\forall[\downarrow \vee \uparrow]$     | <b>weak</b> (in restriction) | ever, any                                            | ooit, enig                                        |
| anti-additive                                         | $f(A \cup B) = f(A) \cap f(B)$                                    | $\forall[\downarrow \vee \downarrow]$   | <b>medium/strong</b>         | lift a finger, in years, sleep a wink, budge an inch | ook maar één, in jaren, met zijn vingers aanraken |
| additive & multiplicative (copula “is”)               | $f(A \cup B) = f(A) \cup f(B)$ and $f(A \cap B) = f(A) \cap f(B)$ | $\uparrow \uparrow$ (no quantifier)     | none                         | —                                                    | —                                                 |
| anti-additive & anti-multiplicative (negative copula) | $f(A \cup B) = f(A) \cap f(B)$ and $f(A \cap B) = f(A) \cup f(B)$ | $\downarrow \downarrow$ (no quantifier) | <b>superstrong</b>           | give a damn, need (modal), either, at all            | hoeven, ook maar, geen moer, voor de poes         |

As I said, I’m not endorsing this taxonomy. Indeed, as we will see, I have considerable reservations about it. But my first point is that the polarity-based account of quantificational structures is just as fine-grained as the model-theoretic account and shows the natural syntactic reflexes of the key properties drawn on to support Zwart’s taxonomy of NPI markers, were that taxonomy viable. I do have some observations about the data, which I believe the proof-theoretic account can cast in a new, and perhaps more explanatory light.

### Three operations on inferential depth

The first thing to note is that weak determiners all appear to be involved in some form of range coverage – inferential breadth, if you will. ‘Ever’ in ‘I haven’t ever seen that’ signals that the breadth of the temporal domain is expansive, that the  $\downarrow$  inference is applicable across a broad range of times (nominally, without restriction). ‘Any’ in ‘I didn’t see any student’ similarly opens the breadth of the individual domain; the inference ranges freely across a broad range of students (nominally, all of them).

The strong determiners, on the other hand, seem to be involved in signaling a commitment to *depth* of inference. It drives inferences to the lower end of the relevant scale. Once the range is expanded, the scalar minimizer identifies a low end of a scale within that open range, signaling that the  $\downarrow$  inference is being taken all the way down to what is nominally the minimal unit. ‘Budge an inch’ doesn’t just say the motion domain is open, it says the inference reaches the (nominally) smallest possible unit of motion. ‘Sleep a wink’ says the inference reaches the (nominally) smallest possible unit of sleep. ‘Lift a finger’ says it reaches the (nominal) minimal unit of effortful action. I say “nominal” in all these cases because, as noted earlier, inferential breadth and depth are never exhausted in these cases. In point of fact, there is no established broadest range of some predicate nor a terminal depth of inference. But we can draw on these

expressions to signal that we are serious about these downward entailing inferences and will take them as far as practical.

However, as I noted, I'm not entirely sold on the taxonomy. Notice that anti-additivity doesn't seem to be necessary to introduce strong determiners. They seem to be able to appear within weak NPI positions, and are perhaps even improved when they co-occur with so-called weak NPIs. Thus, we have examples like the following:

- ?Everyone who has helped a bit will be rewarded.
- Everyone who has ever helped a bit will be rewarded
- ?Everyone who gave a damn will be rewarded
- Everyone who ever even a little bit gave a damn will be rewarded

These judgments are, in my opinion, weak. Even the contrastive differences are weak. This is just an observation that the frequent appearance of strong NPIs in anti-additive environments may not be related to the anti-additivity of the environment. One thing that is clear is that these NPIs are up to different tasks in establishing that we are serious about pursuing downward inferences: Weak NPIs commit us to a broad range of things under discussion; the Strong NPI demands that we take the inference to a very deep level of granularity. And this takes us the matter of the so-called superstrong NPIs.

## **Fruitfulness and the Upper Left Periphery**

The analysis developed in the preceding sections treated polarity markers as guides to inferential direction and depth.  $\downarrow$  marks the direction in which inferences run through a domain, and the depth of propagation varies from shallow conventional marking to the explicit bottom-pointing of scalar minimizers and (nominal) ontological floor markers. This account handles the weak and strong NPI cases cleanly. But a class of expressions that have traditionally been grouped with NPIs turns out, on closer examination, to be doing something categorically different; they are not marking inferential direction or depth for inferences on the core predicates but assessing what I will call the *proof-theoretic fruitfulness* of the proposition on offer. And *fruitfulness*, it turns out, is also not a binary property but is also a scalar one, with a rich and structured vocabulary for locating propositions at different points on the fruitfulness scale. Before we get to the scale, we should say a bit about this idea of fruitfulness and the role it can play in a proof-theoretic semantics.

## **From Fregean Assertion to the Assertion of Proof-theoretic fruitfulness**

Proof-theoretic fruitfulness connects indirectly to an idea in Frege's *Begriffsschrift* that is often overlooked in the semantics of natural language. Frege's judgment stroke  $\vdash$  marks the transition from merely entertaining a content to putting it forward – committing to it as available for inferential use. Frege was careful to insist that  $\vdash$  does not predicate truth of the content; it marks an act, the act of assertion, which makes the content available as a premise in subsequent derivations.<sup>10</sup> What Frege's system lacks (and what the fruitfulness cases show is needed) is a

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<sup>10</sup> For discussion, see Ruffino (2022, chapter 11), and Ruffino, San Mauro, and Venturi (2021).

separate marker for proof-theoretic fruitfulness. That is, it is not enough to signal that the speaker is committing to the content as true, but we need a signal that the content is worth committing to at all, whether it opens derivational paths that are valuable to the current proof.

Let's call this marker  $\vdash_f$ . Where  $\vdash$  puts a proposition forward as available for inferential use,  $\vdash_f$  assesses whether that inferential use is worth pursuing, and to what degree. 'I don't give a damn' is  $\vdash_f$  with  $\downarrow$  at the extreme negative end of the fruitfulness scale – a total fruitfulness rejection. 'That's crucial' is  $\vdash_f$  at the extreme positive end. 'I'll keep that on the back burner' is  $\vdash_f$  near zero from above. The fruitfulness operator is scalar throughout, admitting of the full range of degree modification that natural language provides for it.

Frege could not have  $\vdash_f$  because his entire project required context-independence; asserting P is asserting P, full stop, regardless of what proof is in progress. Fruitfulness, on the other hand, is essentially context dependent. A hypothesis is fruitful relative to a particular derivation – that is, relative to what is being proved. But in a proof-theoretic semantics of natural language, context-dependence is not a defect; it is business as usual. Proofs involve goals and strategies for achieving those goals. Some premises help you get there, and some, even if true, do not appear to be productive. Notice that this isn't the same as relevance. Something can be quite relevant, but not useful to the proof-theoretic task at hand.<sup>11</sup>

The standard fruitfulness rejections ('I don't give a damn', 'I don't give a fuck', and my favorite example from Antioquia, Colombia, 'me importa un culo'), are superstrong NPIs, licensed only in antimorphic contexts. But this classification obscures what they are actually doing. They are not marking inferential direction for the predicates. They are making assessments of fruitfulness, placing those assessments at the extreme negative end of the scale. The idiomatic fixity of these expressions suggests that they are towards full negative end of the scale.

Natural language has elaborate resources for grading the fruitfulness of propositions relative to an ongoing inferential exchange. At the high end of the scale, expressions like 'crucially', 'critically', and 'importantly' mark that a proposition is highly fruitful – that it opens essential derivational paths, that the proof in progress cannot proceed without it. Further down, 'usefully', 'interestingly', and 'notably' mark moderate fruitfulness – the proposition opens paths worth pursuing but not essential ones. Lower still, hedged expressions like 'I'll consider that' and 'I'll keep that on the back burner' mark near-zero fruitfulness. The proposition is being acknowledged as potentially useful, but no commitment is being made to pursue it now. 'That isn't helpful' falls below the threshold of fruitfulness entirely – the proposition fails to open any useful paths.

At zero and below, the scale continues into negative fruitfulness; these are propositions that not only fail to open useful paths but actively disrupt the proof in progress. 'That's unhelpful' marks weak negative fruitfulness. 'That's nonsense' marks stronger negative fruitfulness. And towards

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<sup>11</sup> I understand that it is possible to "fine tune" relevance so that it signals something that is not merely relevant but likely to be fruitful. But I take the normal sense of 'relevance' to merely signal something that is on topic or applicable. Of course, I am on record as supporting meaning modulation, so the meaning of 'relevant' can be morphed to signal fruitfulness, if one desires. It does seem to me, however, that the squishiness of 'relevant' can lead to some abuses. To wit: anything can seem "relevant" if the theory is supposed to run through relevance alone.

the bottom of the negative fruitfulness scale is Rhett Butler's, 'I don't give a damn', plus 'I don't give a fuck', and 'me importa un culo'. These are approaching total fruitfulness rejections, suggesting that the proposition has near zero inferential value, that it opens no paths worth pursuing; the speaker is refusing to engage with its inferential potential entirely.

### **The Jespersen Cycle of fruitfulness expressions**

The fruitfulness expressions also show Jespersen Cycle behavior, and in real time. 'I don't give a fuck' appears to be a kind of absolute fruitfulness rejection – total, non-compositional, maximally emphatic. The problem is that the expression is being weakened before our eyes. 'I have zero fucks to give' is already morphing into a quantificational expression. 'Fucks to give' is becoming a count noun, repackaged as a depletable resource. 'I barely give a fuck' is further along still – the expression is becoming fully scalar, admitting of degree modifiers and threshold adverbs. 'I almost give a fuck' places the speaker just below the threshold of inferential engagement. Each step is the Jespersen Cycle in progress: the absolute fruitfulness assessment is being converted into a gradable scalar property, losing its total rejection force as it becomes compositional. The bleaching is not of phonological material, but of absolute fruitfulness – the conversion of a near total proof-theoretic rejection into direction on a graded scale, which is precisely what grammaticalization always does to emphatic expressions.<sup>12</sup>

### **What truth conditions miss**

What this analysis reveals, more broadly, is that thinking about natural language meaning proof-theoretically opens up dimensions of linguistic structure that truth-conditional and model-theoretic frameworks systematically miss. A model-theoretic approach asks what conditions

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<sup>12</sup> The discussion of fruitfulness raises the question of whether it is just a way of describing how we carry out proof theory or whether it is part of the grammar itself. Presumably it is both. If this is right, then the question arises as to where such elements reside in the grammar. This is a topic for another paper, but for those who are interested in linguistic cartography in the sense of Cinque (1999, 2006) and Rizzi (1997, 2004) rough idea comes to this: The fruitfulness cases suggest that the left periphery of the clause extends upward into a discourse-evaluation zone above Rizzi's (1997) ForceP. The natural ordering of discourse-evaluating adverbials reveals a hierarchy of functional heads: At the highest level, RelevanceP ('incidentally', 'by the way', 'apropos') marks whether the incoming content is connected to the current discourse at all. This is the precondition for all lower assessments.

Below that, SincerityP ('frankly', 'honestly', 'candidly') marks the speaker's communicative disposition, signaling that the content is being offered without social filtering or hedging. These are not epistemic markers (they say nothing about the strength of the speaker's evidence) but sincerity markers, flagging that the content is being offered in good faith. 'Bullshit!' is a downward entailing rejection of sincerity; it marks that the content is being offered in bad faith, introducing false paths into the proof rather than genuine ones.

Below that, FruitfulnessP ('critically', 'importantly', 'crucially', 'significantly', 'usefully', 'interestingly', etc.) marks the inferential value of the content at various degrees, asserting that it opens derivational paths worth pursuing. The negative end of FruitfulnessP shades into the superstrong NPI territory: 'I don't give a damn', 'I don't give a fuck', etc, which are located in FruitfulnessP, marked with ↓, and indicating the extreme negative end of the scale. Below that, we would find Rizzi's standard left periphery handling Force, Topic, Focus, and Finiteness.

must hold for a proposition to be true. A proof-theoretic approach asks what role a proposition plays in a derivation – what it contributes to the ongoing inferential project, how deep the inferences it licenses run, and whether it is worth incorporating into the proof at all. The distinction between inferential direction, inferential depth, and inferential fruitfulness (between  $\downarrow$ , depth markers, and  $\vdash_f$ ) is invisible from a truth-conditional perspective because truth conditions are a property of propositions in isolation, not of their role in derivations. It becomes visible only when we ask the proof-theoretic question: what is this expression doing in the current inferential project?

The polarity marker framework, by treating  $\downarrow$  as a proof-theoretic primitive rather than a truth-functional operator, makes these distinctions available for linguistic analysis for the first time. The NPI taxonomy that results is not a taxonomy of algebraic properties of licensing environments – it is a taxonomy of the different jobs that expressions can do in a proof-theoretic exchange: marking direction, marking depth, and marking fruitfulness. These are three distinct contributions to the inferential structure of discourse, and natural language has distinct grammatical resources for each of them. Standard NPI theory, by treating all three as variants of the same negative polarity phenomenon, has been grouping together things that are fundamentally different in kind, and the difference only becomes visible when we stop asking what makes a context negative and start asking what role an expression plays in a proof.

I’m not here to “solve” the problem of NPI licensing, assuming there is a solution. My goal here is only to provide an alternative framework from which to view the problem space. In that alternative framework, there is no interesting difference between so-called negations and NPIs. Both are fundamentally just  $\downarrow$  polarity markers. This of course opens up questions about the variation available among  $\downarrow$  markers, and the contextual signals as to how far the inference path the speaker is committed to. In some cases, it will be towards exhaustion. In others, it won’t be particularly deep. The literature on NPIs is extensive and dense, and, as I said, I am not here to solve it, whatever that might mean. I do believe, however, that once we start view the topic in terms of variations on  $\downarrow$  marking and inferential depth, we will have a more fine-grained toolkit with which to investigate these matters.

### 3.3 Dowty and Negative Concord as Polarity Marking

The most direct anticipation of the present proposal comes from Dowty (1994), who argued that negative concord marking is polarity marking rather than semantic negation. The phenomenon at issue is negative concord – the appearance of multiple negative elements expressing a single negation in languages like Italian, Spanish, and many others:

- Italian: ‘Non ho visto nessuno’ – (= “I didn’t see anybody”)
- English negative concord dialect: ‘I didn’t see nobody’

The prescriptive analysis of standard English says this is a double negation (two  $\neg$  operators that cancel each other), but this of course predicts the wrong interpretation in negative concord languages, where the result is a single negation, not affirmation. The standard response is to say that in negative concord languages n-words like ‘nessuno’ are not negative quantifiers but rather

negative polarity items that require a licensing negation – but this raises the question of what they contribute semantically if not negation.

Dowty's proposal is that negative concord markers are polarity markers that signal downward entailing environments. They carry a negative polarity feature that participates in monotonicity reasoning without introducing a semantic negation operator. The negative concord words in such cases are not contributing the classical negation  $\neg$  to the logical form; they are contributing a  $\downarrow$  marker that interacts with the inference rules of the surrounding context. (This is given a rigorous formal framework for natural logic in Moss (2012).

Consider 'Nobody had no fun' in an English negative concord dialect. On the classical double-negation analysis, this would mean 'Everyone had some fun' – the two negations cancel. On Dowty's polarity marking analysis, what's happening is:

- 'Nobody' introduces  $\neg\downarrow\text{person}(x) \vee \downarrow\text{fun}(e)$
- 'no fun' is a voiced  $\downarrow$  marker on 'fun' — making explicit a polarity that the derivation already knew to be present from the first 'no'.

There is no double negation because there was never a second negation. The lower 'no' in 'no fun' is not introducing a second negation that cancels the first; it is voicing a  $\downarrow$  marker that was already present in the inferential context established by 'no one'. Negative concord languages are simply languages that voice their  $\downarrow$  markers more promiscuously than standard English does. In standard English, the  $\downarrow$  marker introduced by 'no one' is sufficient and the scope position remains unmarked; in negative concord languages, the  $\downarrow$  marker is lexicalized as another n-word.

Where I part company with Dowty is on the idea that there are any negations at all. What he did for negative polarity items, he could just as easily have done for negation itself, at least as far as the semantics is concerned (whether n-words carry other grammatical responsibilities is another question). Interestingly, L&Z criticize Dowty for the redundancy in his proposal: He tags negations and quantifier positions with  $\downarrow$  markers. Why be redundant? But what L&Z failed to see was that the resolution of the redundancy by opting for classical negation was not the only option.  $\downarrow$  markers are all you need for quantifier positions, n-words, and NPIs.

One final note on Dowty's idea that the negative polarity item was marking a downward entailing environment. On Dowty's view, this marking was to assist in the inferential process of our natural logic. However, subsequent psycholinguistic research by Szabolcsi, Bott, and McElree (2008), shows that such markings don't speed up our inferential processing in these contexts. One explanation offered by Szabolcsi et al is that these markings are not merely indicating an inferential direction but are calling for a deeper inferential process – a more "exhaustive" inferential path. Earlier, I endorsed a version of this idea, but I cautioned against pushing depth of inference to inferential exhaustion. This is a genuine difference, as we will see in the next sub-section, in which I discuss Chierchia's "exhaustifier" operator.

### 3.4 Chierchia on “Nessuno”<sup>13</sup>

A revealing test case for the polarity marker proposal comes from Chierchia's (2006, 2013) treatment of negative concord in Italian. His principal example is one we drew on earlier: *non ho visto nessuno* — in English, “I didn't see nobody,” or in standard English, “I didn't see anybody.”

Chierchia's analysis assigns *nessuno* the semantics of an existential quantifier over a domain *D* of persons — it is a negative concord item (NCI) that introduces a free choice or domain-widened existential, not a semantic negation. This analysis already takes a significant step away from the classical picture: *nessuno* is not contributing  $\neg$  to the logical form. But it immediately creates a dilemma for any account that retains classical negation as a primitive elsewhere in the derivation.

The problem is that *nessuno* as a bare existential is too weak.  $\exists x \in D[\text{person}(x) \wedge \text{saw}(I, x)]$  just says I saw someone — the wrong reading entirely. And negating it straightforwardly with classical  $\neg$  gives  $\neg \exists x \in D[\text{person}(x) \wedge \text{saw}(I, x)]$ , which is “I didn't see anyone” — closer, but it leaves open all the sub-domain alternatives: I might have seen all the professors, or John and Mary, or just John, as long as I didn't see every person in *D*. The account needs to exclude those sub-domain alternatives to derive the intended reading “I saw nobody.”

Chierchia's solution is to introduce the exhaustifier *O\_ALT*, which is a covert operator that scopes above the entire sentence and excludes all the sub-domain alternatives:

— *O\_ALT* [ (pro) non [ho visto nessuno[[n-D]]] ]

*O\_ALT* takes the negated existential and strengthens it by asserting that none of the sub-domain existentials hold either — I didn't see the professors in the domain, nor John and Mary, nor even the singleton set of John. The result is the intended reading “I saw nobody.”

Using *O\_ALT* is a solution, but it is a solution to a problem that only exists because classical  $\neg$  is the wrong primitive. On the polarity marker picture, the dilemma dissolves immediately. *Nessuno* introduces  $\downarrow$  with “person.” *Non* is a voiced  $\downarrow$  marker with “saw.” The logical form is simply:

—  $\downarrow \text{saw}(I, x) \vee \downarrow \text{person}(x)$

with *x* a free variable. Now consider what each  $\downarrow$  is doing. The  $\downarrow$  on “person” does not mean “non-person.” It means that the restriction carries downward entailing inferential behavior. Inferences go from the full domain of persons down to subsets of persons, and so on. The restriction remains about persons; only the inferential direction is marked. There is no complement of the person domain constructed; there is only the understanding that because it is a downward polarity environment, you can substitute progressively smaller subsets of the set of persons.

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<sup>13</sup> Thanks to Anna Szabolcsi for pointing out errors in a previous draft of this sub-section.

Similarly, the  $\downarrow$  on "see" means that the seeing is in a downward entailing context. And because  $\downarrow$  is defined as downward closure, it propagates potentially through all sub-domains of seeing events, down to some contextual limit. For example, if you didn't see anybody, you didn't see anybody while using opera glasses.

The logical structure in Chierchia's system utilizes two separate logical operators doing two different jobs:  $O\_ALT$  and  $\neg$ . On the polarity marker picture this collapses to a single, simpler operator that still manages to do the work of Chierchia's two operators. One  $\downarrow$  marker is doing both jobs at once, not because  $\downarrow$  is doing more work, but because  $\neg$  was doing too much and too little. Classical  $\neg$  aggressively flips truth values, but that is all it does; it does not propagate downward through the inference structure. To put it another way,  $O\_ALT$  was needed to supply the downward propagation that  $\neg$  lacked. But  $\downarrow$  is defined as downward entailment toward closure; the propagation is built in.  $O\_ALT$  is a patch for the inadequacy of  $\neg$  as a primitive. Once  $\neg$  is replaced by  $\downarrow$ , the patch is unnecessary.

One might object that Chierchia's exhaustifier ensures inferential exhaustion, which  $\downarrow$  does not. But it does seem to me that while *nessuno* signals some aggressive downward entailment, it is far from exhaustive in typical usage. "I didn't see nobody last night" remains felicitous even if the speaker passed a stranger in the street. And this is true even if domain restriction is in play. "A: I didn't see nobody at the club last night. B: Weren't Sam and Pat there? A: Yah but no one else was there; it was too crowded."<sup>12</sup>

### 3.5 Horn and the Copular Analysis

The most historically grounded challenge to the classical theory of negation, at least in its syntactic form, comes from Horn (1989). Drawing on a rich body of cross-linguistic and historical evidence, Horn argues that the standard treatment of 'not' as a sentential operator that applies to a proposition and yields its truth-functional complement misrepresents the actual structure of natural language negation. The primary form of negation in natural language is not  $\neg P$ , but 'S is not P' – a predication with a negative copula. This is the Aristotelian picture: negation attaches to the copula rather than to the proposition as a whole, as in ' $\neg$  [S is P]'.

Horn's evidence for this is broad. Negation in natural language is not a freely applicable propositional operator. It has restricted distribution, it interacts asymmetrically with scope, and it behaves in ways that model-theoretic treatments of  $\neg$  have difficulty capturing. The cross-linguistic picture supports the copular analysis: in language after language, what looks like sentential negation turns out on closer examination to be predication with a negative marker on the copula or verbal complex. One even sees this with the way that classical negation is rendered in natural language: 'it is not the case that...'. Notice that a negative copula has been smuggled into this construction.

As I understand Horn, however, the negative marker is a classical negation. There is, on Horn's view, a scale in play, and the negation flips that scale, but it is still, for all that, still a classical negation. This fusion of classical negation with negative copulas raises questions. What does it mean to fuse a negation with the copula? (I actually wondered about this when I learned

Aristotelian logic, assuming at the time that post-Boolean negation made sense in that context.) I'm not sure the Boolean negation story has a coherent answer to that question. However the polarity marker framework does offer a credible answer: 'is not' is simply the standard copula tagged with the  $\downarrow$  polarity marker, which will be applied to the following predicate. And the fact that it is just a simple polarity maker may help explain why contraction is so easy – the marker cliticizes with the copula. The difference between 'Socrates is a god' and 'Socrates isn't a god' is not the application of  $\neg$  to a proposition; it is the difference between  $\uparrow\text{god}(s)$  and  $\downarrow\text{god}(s)$  – a difference in the polarity marking of the predication itself. The copula carries the polarity marker and the inferential behavior follows from the marker rather than from any negation operator. Horn was right about the phenomenon and right about the diagnosis; the polarity marker framework provides an alternative mechanism that is, in my opinion, better aligned with his theory tout court.

We can also show how this idea works in a medieval logic that has reduced the Aristotelian syllogism into two basic rules: the dictum de omni and dictum de nullo, as described earlier. Consider the following syllogism:

- (1) *No god is mortal.*
- (2) *Socrates is mortal.*
- (C) *Therefore, Socrates is not a god.*

In the polarity marker framework, the derivation proceeds as follows:

**Premises:**

- 1. No god is mortal:  $\downarrow\text{god}(x) \vee \downarrow\text{mortal}(x)$
- 2. Socrates is mortal:  $\uparrow\text{mortal}(s)$

**Conclusion:**

$\downarrow\text{god}(s)$  – from 1 and 2 by dictum de omni. "Socrates is not a god."

The conclusion  $\downarrow\text{god}(s)$  is the formal representation of 'Socrates is not a god'. The 'is not' in the conclusion is simply the surface realization of the  $\downarrow$  marker on the predicate 'god' as it applies to Socrates. No negation operator appears anywhere in the derivation. The  $\downarrow$  marker was introduced by 'no' in premise 1. Premise 1 is also a way of stating that all mortals are  $\downarrow$ -gods. Or to put it another way,  $\downarrow$ -gods are a superset of the mortals. Thus ' $\downarrow\text{god}$ ' can replace 'mortal' in a positive polarity environment by *dictum de omni*.

Just to be clear, the thesis I am advancing is that the polarity marking proposal is continuous with Horn's project. And quite possibly, it would be the natural way to understand the negative copula before the advent of Boolean negation.

### 3.5 Schein's Noughty Bits

Schein (2016) offered an analysis of tense that, on the face of it, appears to be a defense of classical negation, because it aims at rescuing the law of excluded middle. The starting point is the contrast between the following cases:

- (1) The mirrors are smooth.
- (2) The mirrors are not smooth.
- (3) The mirrors circle the telescope's reflector.
- (4) The mirrors do not circle the telescope's reflector.

Sentences (1) and (2), involving a homogeneous predicate, appear to violate the Principle of Excluded Middle: there is a reading on which (1) is neither simply true nor simply false (some mirrors are smooth, some aren't), yet (2) seems to have a similar intermediate status rather than being simply the negation of (1). Sentences (3) and (4) behave classically; they are straightforward contradictories. The *standard* logical forms  $\text{Smooth}(m)$  and  $\neg\text{Smooth}(m)$ ,  $\text{Circle}(m)$  and  $\neg\text{Circle}(m)$ , predict no difference between the two pairs. Yet the difference is robust and systematic.

Schein's central claim is not that the usual sentential treatment of negation is too coarse to capture how negation actually works in the presence of eventive structure. Schein argues that in natural language, 'not' should not be treated as a plain sentential connective. Instead, it functions as an adverb of quantification (what he calls *noughtly*) paired with event quantification rather than merely attaching to a completed proposition.

Schein's diagnosis is not that 'not' itself is ambiguous or nonclassical. Rather, the apparent failures of Excluded Middle arise because one has suppressed the relevant event structure and mistaken a framed event-semantic configuration for a simple propositional negation. His proposal is explicitly intended to resolve these puzzles within classical logic, without ambiguity in the meaning of 'not' and without appeal to negative events as part of the ontology.

Accordingly, Schein's proposal is that negation is built into restricted event quantification itself, in forms of the type  $[\exists E: \Phi[E]] [\text{No } e: Ee] \Psi[e]$ , where a framing event or region  $E$  delimits the relevant domain and the negative force is carried by the restricted quantifier over subevents. On this view, 'not' is not understood as an external propositional operator, but as negative quantifier expression ('no'), which is part of the internal architecture of the event quantification.

You may have guessed where I am going next, because with this move Schein has set the ball on the tee for the analysis of quantification proposed in Section 2. 'No' is simply a generalized quantifier with the polarity structure  $\downarrow A(x) \vee \downarrow B(x)$ , which in this case, takes the form:  $\downarrow E(e) \vee \downarrow \Psi(e)$ .

Here is another way to put the point. If the real semantic action lies not in a sentential connective but in the structured distribution of negative force across a restricted event domain, then Schein's forms can be re-read as evidence that natural-language negation is already sub-propositional in character. On that reinterpretation, the negative component in a form like  $[\text{No } e: Ee] \Psi[e]$  can be modeled not as an irreducible  $\neg$ , but as a polarity configuration – roughly, a  $\downarrow$ -marking on the event-domain condition together with a corresponding  $\downarrow$ -marking on the event predicate. In that sense, Schein's architecture would be tracking something real: negation belongs with the fine structure of event individuation and restriction, not with a monolithic propositional operator. As

with Horn's account of copular negation, he is so close yet so far. The only thing he had to do was replace the alleged classical negation with the polarity marker.

### 3.6 Zeijlstra on [uNEG] and the feature [iNEG]

Zeijlstra (2022) develops one of the most systematic and technically sophisticated treatments of negation and negative dependencies in recent literature. His central proposal is that the negative morphology on n-words is syntactic agreement rather than semantic negation. N-words in negative concord languages carry an uninterpretable negative feature [uNEG] that needs to be checked against an interpretable negative feature [iNEG] on a dedicated negative head in the clause structure. The articulated n-words don't themselves contribute semantic negation; they agree with a negative head that does. This is a crucial step toward the polarity marker proposal. Zeijlstra has pulled apart the surface form (the negative morphology on n-words) from the logical primitive (the semantic negation operator). This separation is exactly what the polarity marker framework requires.

But Zeijlstra stops short of the full dissolution in one crucial respect. He retains the view that somewhere in the clause there must be an element carrying [iNEG], which is an interpretable negative feature that introduces classical Boolean negation plus existential closure over free variables. In Strict negative concord (NC) languages, this element is a covert abstract operator  $\text{Op}_{\neg}$  sitting in the Spec of NegP. In Non-Strict NC languages, it is the overt negative marker itself (e.t. Italian 'non', Spanish 'no'), base-generated in Neg<sup>o</sup>. Either way, classical  $\neg$  survives in his system, relocated from the surface morphology to a more abstract position and in a more abstract form, but not eliminated.

The polarity marker proposal says this residual  $\neg$  is unnecessary. Everything  $\text{Op}_{\neg}$  was posited to do can be done by  $\downarrow$  without invoking the truth-functional complement operation. Before pressing this argument, however, it is worth examining Zeijlstra's syntactic architecture in detail. As we will see, two problems arise from within his own framework that suggest his syntactic story is less secure than his feature-theoretic apparatus implies.

#### The interpretation timing problem

The first problem arises with Zeijlstra's treatment of neg-raising. Sentences like Italian 'non credo che ha fatto niente' ('I believe he didn't do anything') where 'non' is in the matrix clause but the NC reading pairs it with 'niente' in the embedded clause, look like NC crossing a clause boundary. Zeijlstra's response is that this is neg-raising: 'non' is phonologically in the matrix clause but semantically interpreted in the embedded clause, so the Agree relation between 'non' and 'niente' is actually local.

One problem is that this looks a lot like Italian clitic climbing. If that is right – if 'non' is behaving like a clitic and cliticizing to 'believe', then the result is something on the order of 'believe-not...' rather than 'it is not the case that I believe...'. The second problem is that this move runs counter to the standard picture of when interpretation happens. The entire motivation for LF as a level of representation (and for Quantifier Raising as an operation) is that

interpretation follows movement, not precedes it. QR moves a quantifier to a position from which it takes scope, and interpretation happens at that derived position. Zeijlstra's neg-raising account requires 'non' to be interpreted before it moves, which inverts this logic entirely.

### **The learnability problem**

Zeijlstra offers an acquisition account in which the learner assigns [uNEG] to a negative element when the input contains multiple negative morphemes corresponding to a single semantic negation, and [iNEG] when the negative element is the sole contributor of negation. The distinction between NC and Double Negation languages like standard English is then a parametric difference in which feature value the learner assigns to negative elements, and the Jespersen Cycle is the diachronic history of those feature values shifting as the phonological salience of negative markers changes.

The problem is that this account mischaracterizes what is being learned. If  $Op^-$  is present in every language (sometimes voiced, sometimes covert), then the learner is not acquiring knowledge of whether to have classical negation as a syntactic category. They are acquiring something much more mundane – i.e., whether their language voices the negative operator overtly in a given morphosyntactic environment. That is not a deep parametric difference in logical structure; it is a fact about morphophonological conventions of the same kind as whether a language voices its tense morphology on the verb or drops it. The Jespersen Cycle on this view is simply the history of a voicing convention, not the history of whether classical negation is grammaticalized as a syntactic category. The [uNEG]/[iNEG] distinction loses most of its theoretical weight once we recognize that the  $\downarrow$  operator is always present in every language and the only question is whether it gets phonological content in a given morphosyntactic environment.

Independent empirical pressure on the [uNEG]/[iNEG] distinction comes from an observation by Szabolcsi (personal communication). Zeijlstra's typology requires the distinction between Strict and Non-Strict NC to be a stable, language-level parameter: a given language's sentential negation morpheme either carries [iNEG] (making it semantically negative and the local instantiation of  $Op^-$ ), or [uNEG] (making it a syntactic signal dependent on a covert  $Op^-$ ) for its licensing. The typology is binary, and individual languages are sorted into one category or the other.

Szabolcsi points out that this binary typology does not hold. As Surányi (2006) first observed and Szabolcsi (2018) subsequently analyzed in detail, Hungarian has a hybrid system in which both Strict and Non-Strict NC patterns are available within the same language. Crucially, both types can appear within the same sentence! Catalan presents a parallel case, as documented by Espinal et. al (2016). If a single language can deploy both patterns, and if both can co-occur in a single derivation, then the [uNEG]/[iNEG] distinction cannot be a lexical property stably assigned to the sentential negation morpheme across the language. The morpheme would have to carry different feature values in different constructions, which undermines the entire feature-checking architecture. Agree requires stable feature values: if the same morpheme is [iNEG] in one environment and [uNEG] in another, there is no determinate feature for  $Op^-$  to check against, and the derivation has no principled way to proceed.

This empirical observation confirms the theoretical point made above from a different direction. The problem is not merely that Zeijlstra's acquisition story misdescribes what learners are acquiring. The problem is that the object the learner is supposed to be acquiring – a fixed [iNEG] or [uNEG] value for the sentential negation morpheme – does not exist as a stable property of the morpheme in the first place. What the Hungarian and Catalan data show is that the relevant variation is not between languages but within them, and that it is sensitive to constructional and contextual factors that Zeijlstra's binary parameter cannot capture. Notably, Szabolcsi (2018) herself attempts a unified analysis of the Hungarian hybrid system within Zeijlstra's general framework, combining his treatment of Strict NC with Chierchia's (2013) treatment of Non-Strict NC. That two separate theoretical frameworks are required to handle the two patterns within a single language is itself an indication that something has gone wrong at the level of the typological primitives.

The polarity marker framework handles this more naturally. On the  $\downarrow$  account, [uNEG]/[iNEG] fundamentally signal the same operator ( $\downarrow$ ) in different syntactic positions. When there is a difference, it is because the morphology of one of the operators is signaling a deeper inferential strategy. The apparent difference between Strict and Non-Strict NC within a single language is simply a difference in where  $\downarrow$  is anchored in specific constructions; it is a fact about syntactic position rather than a fact about lexical feature values. The Hungarian and Catalan data are not a problem for this account; they are exactly what it predicts.

### **The deeper problem: $\text{Op}\neg$ as the last refuge of classical negation**

These internal problems suggest that Zeijlstra's syntactic architecture is less stable than it appears. But the deeper issue is what  $\text{Op}\neg$  is doing in the first place.  $\text{Op}\neg$  is defined as  $\neg(\exists)$ , which is Boolean negation plus optional existential closure over free variables in its scope. It is the element in Zeijlstra's system that actually introduces classical  $\neg$  into the semantics. N-words don't contribute it directly, and [uNEG] negative markers don't contribute it.  $\text{Op}\neg$  does. This makes  $\text{Op}\neg$  the last refuge of classical negation in his framework. He has correctly stripped  $\neg$  from the surface morphology, but then reconstructed it at the abstract level.

The polarity marker framework asks: is  $\text{Op}\neg$  doing anything that  $\downarrow$  cannot do without the classical  $\neg$ ? The answer is no.  $\text{Op}\neg$  accomplishes three things in Zeijlstra's derivations: it checks the [uNEG] features of n-words and negative markers via multiple Agree; it permits an existential closure that binds the free variables introduced with n-words (so long as they aren't in the restriction of a quantificational construction); and it contributes Boolean negation to the truth conditions of the sentence. The first two jobs are exactly what  $\downarrow$  does in the polarity marker framework, without any appeal to truth-functional complement. The third job (contributing Boolean negation to truth conditions) is precisely what the polarity marker framework argues is unnecessary. As Section 2 showed in detail, the phenomena that have been taken to require classical  $\neg$  in the truth conditions are better described as the behavior of  $\downarrow$  propagated to a contextually determined depth. Classical  $\neg$  is what  $\downarrow$  looks like when propagated all the way to some hypothetical lowest inferential level. It is an idealization, and even if it is coherent, natural language seldom if ever makes that commitment.

### 3.7 Szabolcsi and Zwarts on complementation and weak islands

I've said throughout this paper that I think that classical negation can be removed from both the syntax and semantics. However, it must be conceded that negation can rear its head in many ways. One way it can arise in semantics is when one calls on the complementation operation. For what is set-theoretic complementation if not a form of negation? After all, the complement of a set of things that are *A* is just all the things that are *not A*. This leads to the question of whether one needs to appeal to complementation, and if so, what that looks like on our proof-theoretic account of non-classical negation. In this sub-section, I want to examine a proposal by Szabolcsi and Zwarts (1993) designed to address weak islands and see the problem (and solution) through the lens of proof theory.

The starting point is the observation that negative islands are weak islands. They block extraction of manner, degree, amount, and frequency *wh*-phrases, but not individual-ranging *wh*-phrases. 'Who didn't John invite?' is perfectly grammatical; \*'How didn't John leave?' is not. Szabolcsi and Zwarts argue that the difference lies in the algebraic structure of the *wh*-phrase's domain. They argue that individual domains are full Boolean algebras in that they support meet, join, and complement. Manner, degree, and frequency domains, on the other hand, are only join semi-lattices – they support union but not complementation. Extraction out of a negative context requires the *wh*-phrase's domain to support complementation, which individual domains do and manner/degree/frequency domains do not. Hence the island effect.

This account appears to invoke classical negation, since *complement* of *A* is often defined as  $\{x : \neg A(x)\}$ . But does the introduction of complementation actually require classical negation? No. The same work can be done without classical negation, but how? One quick solution is simply to say that the complement of *B* is:  $U \setminus \{x : B(x)\}$ . That is, it is the universe minus the set of things that are *B*. No  $\neg$  operator is introduced. The operation is entirely well-defined as long as *U* is well-defined. Now I suppose one could argue that the negation sneaks in somewhere when we talk about the operation that removes the set of things that are *B*, but before we go down that road, there is a better path for proof theorists to follow, since worrying about sets really isn't their business.

In the proof-theoretic approach to natural language semantics, the meaning of a lexical item is not a set – it is a bundle of inference rules, the cuts (in Gentzen's sense) and meaning postulates that license reasoning with that item. With a nod to Quine's (1951) "Two Dogma's of Empiricism," we set aside the question of whether this information is analytic or synthetic. It just happens to be information, tied to a predicate, that is reliable enough to be called upon in a proof. For the lexical item 'cat', the inference bundle probably includes rules like: every calico cat is a cat; every cat is an animal; every animal is a living thing. This is a monotone series of categorical inclusion rules – a chain of upward entailments. 'Cat' sits as a node in this chain, connected to more general predicates above it and more specific predicates below it.

*Complementation as Polarity reversal.*

Forming the complement of 'cat' in the proof-theoretic sense is simply running this inference in the  $\downarrow$  direction. From ' $\downarrow \text{cat}(x)$ ' you get ' $\downarrow \text{calico-cat}(x)$ ': Socrates is not a cat, so Socrates is not a

calico cat. The ↓ marker propagates downward through the inference chain by *dictum de nullo*. You do not get to infer ‘↓animal(x)’, because that would be an upward entailment. This is the proof-theoretic analog of complementation; it is not a new set, but a reversal of inferential direction through an existing order.

But now let’s go back to the key insight by Szabolcsi and Zwart. On their view, manner predicates like ‘gracefully’, or degree predicates like ‘temperature’ did not allow complementation. These do not sit in a monotone series of categorical inclusion rules. There is no inference chain of the form ‘all graceful manners are smooth manners’; manner predicates don’t participate in categorical kind hierarchies in the relevant sense. Similarly, ‘temperature’ is essentially a definition or a scale point; it picks out a region on a scale, but there are no rules of the form ‘all temperature is x’ connecting it to other predicates through categorical inclusion. It is an isolated point on a measure, not a node in a monotone order.

When a predicate lacks this monotone inferential structure, polarity reversal is undefined – not because of a set-theoretic failure, but because there is no order to invert. The ↓ marker has no positive order to flip. The complement is unavailable not in the sense that a set fails to exist, but in the sense that the proof-theoretic operation that would generate it has no ordered inference chain to operate on.

This is the proof-theoretic version of the explanation of negative islands given by Szabolcsi and Zwarts. Movement of a wh-phrase through a ↓-marked context (in this case, passing through a surface negation) requires forming the proof-theoretic complement of the wh-phrase’s predicate – which is to say that it requires running flipping the direction of its inference bundle in the ↓ direction. This works for individual-ranging wh-phrases range over predicates like ‘cat’, ‘person’, ‘dog’ – predicates that sit in monotone categorical inference chains and support polarity reversal. However, manner, degree, and frequency wh-phrases range over predicates that lack this monotone structure. They are scale points or definitions rather than nodes in inference chains, and polarity reversal is therefore unavailable. The extraction fails because the proof-theoretic polarity-flipping operation required cannot be executed.

Szabolcsi and Zwarts were right that the difference between the two cases turns on the availability of complementation. What they did not have was a proof-theoretic account of why complementation is available for some domains and not others. The polarity marker framework provides that account – and it requires no classical negation at any point.

## 4. Conclusion

Let’s revisit the central positive. The claim is not that the words ‘Not’, ‘no’, ‘is not’, ‘nobody’, ‘never’ disappear from natural language. These words remain, and they play important roles in grammar and discourse. What disappears is the idea that they represent classical negation. In the proof theory of natural language, no negation operator is needed because the inferential work that negation was thought to do is fully handled by the ↓ polarity marker.

The introduction rules for  $\downarrow$  are given by the lexical items that create downward-entailing environments. ‘No’ introduces  $\downarrow$  in both restriction and scope positions simultaneously. ‘Every’ introduces  $\downarrow$  in the restriction position only. ‘If...then’ introduces  $\downarrow$  in the antecedent position. ‘Is not’ introduces  $\downarrow$  on the predicate without any governing quantifier. These are simply lexical facts about the polarity profiles of the relevant items.

The Brouwerian intuition is vindicated in a new form. Brouwer was right that classical negation is an inflation, but the deflation does not require constructivism or the executability requirement. The deflation comes from recognizing that what classical logic called “negation” is, in natural language, simply  $\downarrow$ -marking. The semantic anchor for the  $\downarrow$  marker is not Boolean negation but entailment direction: a  $\downarrow$ -marked formula licenses inferences from sets to subsets. This is a perfectly well-defined semantic property that makes no reference to negation.

Horn is vindicated, too. ‘Is not’ is a copula; specifically, it is the  $\downarrow$ -marked form of the copula, just as “is” is the  $\uparrow$ -marked form. The difference between them is not operator versus non-operator; it is a difference in polarity marking within the structure of predication itself. Griss’s program of negationless mathematics turns out to have a natural analog in natural language, but the analog is not impoverished in the way Griss’s mathematics threatened to be. Natural language is full of “negation”: the n-words are all still there. What is negationless is the semantics and the proof theory. The derivation system for natural language inference needs no  $\neg$  operator, no negation rule, no definition of downward entailment in terms of negation. It needs only  $\uparrow$  and  $\downarrow$ ,  $\wedge$  and  $\vee$ , and (if Ludlow and Zivanovic (2022) are right) the two medieval inference rules.

The medieval logicians, working without the resources of modern logic, constructed a theory of inference from the ground up based on the structure of predication and the copular distinction between ‘is’ and ‘is not’. They did not need the operator  $\neg$  because they were not trying to model inference as truth-functional computation. They were tracking something in natural language that modern formal logic, with its Boolean apparatus, has obscured. The polarity marker framework restores the insight: inference in natural language semantics is fundamentally about the direction of entailment, and that direction is encoded in the polarity of predications. Negation, as a logical primitive, was always a detour for semantics – certainly for proof-theoretic semantics. But once we see this, we have to question attempts to read classical negation into the *syntax* of natural language. For sure, there are so-called n-words. The question is, what are they representing? Some of them are surely signaling polarity markers. Often, they are calling for depth of inference (if not exactly inferential exhaustion). None of them, I hypothesize, are standing in for classical negation.

As I noted, this is a working hypothesis rather than a universal law. There is no end of natural language constructions in which classical negation has been proposed as a part of the analysis. I have only scratched the surface in this paper. What I can say with some confidence is that I have yet to see a construction in which the polarity analysis cannot replace the classical negation analysis, and I have yet to see a case in which the resulting polarity-based analysis did not deepen our understanding of the phenomenon under investigation.

## Appendix: On the Philosophical Mythologies of Negation

The argument of this paper has not only been technical, involving proof-theoretic semantics, polarity markers, and the dissolution of quantificational structure, but it has also been limited to the syntax and semantics of natural language. I haven't made any claims about the use of negation in the foundations of mathematics, nor even in normative (non-naturalist) logic. But what happens if we take these claims about negation seriously as a philosophical position?

There are reasons we might do this. One might think that the proof-theoretic approach is naturally attached to a kind of idealist project (as Dummett believed). Alternatively, as I have argued in some of my metaphysical work (e.g. *Semantics Tense and Time: An Essay in the Metaphysics of Natural Language*), the semantics of natural language opens a window onto the structure of reality. This is not the place to defend such ideas. The question for now is this: What happens if this picture of "negation" in the semantics of natural language is also illuminating of the metaphysics of "negation." What if there is no such thing as negation? If there is only polarity and inferential direction?

It is a fascinating question. Negation (what people think is negation) has been one of the great obsessions of 20<sup>th</sup> Century philosophy, and much of what philosophers have said about it turns out, on examination, to rest on an uncritical acceptance of classical logic's picture of what negation is. Once we see that picture for what it is (a theoretical choice, not an empirical discovery) a remarkable amount of philosophical mythology begins to dissolve.

### The classical picture and its philosophical shadow

As we noted in the introduction, classical negation is a bivalent, truth-functional operation:  $\neg P$  is true if and only if  $P$  is false, and there is no third possibility. This is not an innocent technical stipulation. It carries with it a metaphysical picture of the world as divided into positive and negative facts and of reality as structured by complementation. When philosophers reach for negation as a philosophical concept, as they have done with remarkable frequency and confidence, they are typically drawing on this classical picture, often without acknowledging that it is a picture at all.

The argument of this essay has been that natural language does not work this way. What we call negation in natural language is not the classical complement operation. It is rather  $\downarrow$  – a marker of inferential direction, a signal that reasoning runs from sets to subsets, from higher-ranked to lower-ranked positions in some ordering.  $\downarrow$  never definitively bottoms out; it propagates to a contextually determined depth and stops there. The commitment is to inferential direction, not to the full truth-functional complement. Classical  $\neg$  is what  $\downarrow$  looks like when we idealize it to exhaustion. As I noted earlier, it is an idealization that natural language speakers rarely if ever realize.

This matters for philosophy because the philosophical uses of negation have almost uniformly assumed some version of the classical picture (even if logic itself was being rejected). The

residual Boolean picture has metastasized into even the work of the most anti-formalist Continental philosophers. They have, even while rejecting logical methodology itself, retained negation, and viewed it as absolute, as constitutive of reality, and often as the metaphysical ground of absence and non-being. On the  $\downarrow$  picture, none of this makes sense. Negation is not absolute and it is not constitutive of reality; it is a guide to inference. It does not bottom out in non-being; it simply tells you which direction to go.

## Heidegger and the Nothing

Ironically, the person most in the grip of the classical 20<sup>th</sup> Century picture of negation is Heidegger. I say ironically, because right smack in the middle of his critique of modernity is one of the central pillars of modernity in logic – classical negation. Heidegger's 1929 lecture “What is Metaphysics?” begins with a question that he treats as both profound and unavoidable: what about the Nothing? Das Nichts – the Nothing – is, for Heidegger, not merely the logical complement of being but a fundamental feature of human existence, revealed in the mood of anxiety. In anxiety, he argues, we encounter the Nothing directly; being as a whole slips away, and what remains is the Nothing itself. The Nothing is not an object or a being, but rather the condition of possibility for the disclosure of being. “The Nothing nothings” – *das Nichts nichtet* – is his famous formulation, intended to capture the active, verbal character of the Nothing as a metaphysical principle.

Carnap's response in “The Overcoming of Metaphysics through the Logical Analysis of Language” (1932) was to treat this as a straightforward grammatical error: Heidegger had taken the logical symbol for negation and turned it into a noun, then predicated activity of it. “The Nothing nothings” is syntactically well-formed but semantically empty; it is a pseudo-statement generated by the misleading surface grammar of natural language.

Carnap was right to be suspicious, but his grammatical diagnosis was sub-optimal. From the perspective of natural language, at least, nouns and operators are easy to verbalize and turn into predicates of action. Operators operate, quantifiers quantify, negations negate. There is nothing incoherent about that. These operations often have outputs: often, allegedly, the empty set, or what we might call “nothing.” That leaves us with a noun, but nouns are easy enough to convert into verbs – we do it all the time (rain, snow, hail), and we are doing it even now (google, text, email), and while you might say that rain doesn't rain or e-mails don't themselves email, microwaves do microwave. So, who knows, once you have nothing at all, maybe it is in the business of doing what nothing does. The question is, do we ever get to nothing at all?

The error in Heidegger is not that he nominalized a logical operator or verbalized what he shouldn't. The deep error is that both Heidegger and Carnap accepted the classical picture of what negation is. Heidegger (perhaps unwittingly) took the core components of classical negation (absolute, exhaustive, truth-functional complement) and drew metaphysical conclusions from it. Carnap accepted the same picture and concluded that the metaphysical conclusions were meaningless. Neither questioned whether the classical picture was the right picture in the first place. They were as far apart as two philosophers could be, yet they shared a flawed assumption.

On the  $\downarrow$  account, the question of the Nothing simply does not arise in the form Heidegger posed it. There is no absolute complement of being, because  $\downarrow$  never produces an absolute complement. When we “negate” a predicate – when we mark it with  $\downarrow$  – we are not gesturing at some metaphysical anti-realm of non-being. We are saying: inferences run downward from here. The commitment is inferential. ‘I’m not hungry’ does not posit the Nothing; it establishes a downward entailing environment in the space of hunger-states, propagated to some contextually determined depth. There is nothing metaphysically deep here, only an inference guide.

The mood of anxiety that Heidegger associates with the alleged encounter with Nothing is real enough as a psychological phenomenon. But the source of that anxiety is misdiagnosed. Something is wrong, but whatever that something is, it isn’t the product of negation.

### **Sartre and the Négatités**

Sartre’s treatment of negation in *Being and Nothingness* (1943) is in some ways more sophisticated than Heidegger’s, but still deeply committed to the classical picture of negation. For Sartre, negation is not merely a logical operation but a fundamental feature of consciousness. Consciousness is defined by its capacity for negation – its ability to put distance between itself and the world, to say no, and to introduce nothingness into the plenum of being. The famous analysis of the café waiter who is “not” his role, the lover who is “not” the beloved, the bad faith that consists in denying what one is – all of this depends on negation being a deep metaphysical capacity of the pour-soi, the for-itself, that the en-soi, the in-itself, lacks.

Sartre introduces the concept of négatités, which are negative realities like absence, lack, distance, and otherness. They function as part of the anti-furniture of the world brought into being by consciousness through its capacity for negation. When I arrive at the café and find that Pierre is not there, I do not merely note the absence of a logical fact. I apprehend a négatité; Pierre’s absence is a real feature of the situation, not reducible to the positive facts. The world is structured by negation because consciousness structures it that way.

Sartre was a careful enough thinker to see that the classical picture of negation had implications he wanted to draw out. But the implications depend entirely on negation being absolute. That is, it depends on  $\neg P$  genuinely excluding  $P$  with no remainder (or, if you prefer, on the complement being fully determinate). On the other hand, if negation is better understood via the operator  $\downarrow$  rather than  $\neg$ , then the négatités are not what Sartre thought they were.

Pierre’s absence, on the  $\downarrow$  account, is not the presence of a negative fact. It is a  $\downarrow$  annotation on the predicate ‘being-in-the-café’ – an inferential marker indicating that reasoning runs downward through that predicate in this context. All that really follows is that Pierre is, depending on how deep you want to carry the inference, not in the bathroom, not in the storage room, not hiding in a crawl space somewhere.

Sartre’s insight is that consciousness involves a peculiar relationship to negation – in particular, that human experience is actively involved in the practice of negation. This much is on point – it is basically an axiom of proof theory as envisioned by Brouwer and Gris, but in a mythologized form. Following negative inferential patterns is something we *do*. It is a cognitive achievement,

and it is directed. It is characterized by the ability to take a distance from what is immediately proffered – things survive the  $\downarrow$  operation. What consciousness does not do is introduce metaphysical nothingness into the world. What it does do is operate with  $\downarrow$ ; it can mark any predicate as downward entailing, can run inferences down any given ordering, can take any positive situation and ask what would have to be different for it to fail. This is a remarkable cognitive capacity. But it does not involve a reified Nothing.

## **Hegel and determinate negation**

Hegel's Logic is pre-“classical,” and doesn't look much like logic after Boole. For Hegel, negation is not the complement of a proposition. The negation of Being is not Nothing in the classical sense but Becoming – a determinate, contentful transition. The dialectical movement of Hegel's Logic depends on negations being productive, which is to say it depends on each negation generating a determinate positive content that drives the dialectic forward.

Hegel was onto something when he insisted that negation in thought is never purely formal and contentless – that when we negate something, we are always negating it in some determinate way, from some determinate standpoint, with some determinate inferential consequences. This is also the case with the  $\downarrow$  picture:  $\downarrow$  is always contextual and always attached to a specific predicate in a specific inferential structure. It is always carrying determinate consequences for how reasoning proceeds.

But Hegel still wanted negation to be a metaphysical engine – the force that drives reality through its contradictions toward synthesis. The *Aufhebung*, the dialectical sublation, depends on negation having an ontological weight that pulls the dialectic forward. On the  $\downarrow$  account, this weight dissolves.  $\downarrow$  is an inference guide, not an ontological force. The dialectic is not driven by the inner negativity of concepts but by the contextual depth to which  $\downarrow$  propagates in a given inferential situation. When the depth of  $\downarrow$  propagation reaches the limits of a conceptual framework and forces a revision, we get something that looks like dialectical movement, but it is a pragmatic and inferential phenomenon, not a metaphysical one. Reality does not have a negative component. Our inferences sometimes are sometimes downward entailing, and they sometimes take us places we did not expect to go.

We can, if we wish, reconstruct the Hegelian project through the lens of polarity-based proof theory, although the result is apt to be wildly offensive to Hegel Scholars. No matter. I proceed despite offending.

First step. We want to view the proof-theoretic project as a shared project, not an individual project. So, we, as a group or community, construct a proof together. This is our proof-theoretic “zeitgeist.” As the proof proceeds, there are theses or ideas that we endorse and which play key roles in our shared inferential project. We mark them as  $\uparrow$ , and we reason from those assumptions. At some point, we arrive at a derivation which looks like a form of inferential failure – an unacceptable or incoherent result. We now flip the polarity on the inferential path that got us there. That path of reasoning is now marked  $\downarrow$ . This is now our version of the Hegelian antithesis. And we have a problem. We have theses marked  $\uparrow$  to which we were strongly committed, and now we end up at a position which we must mark  $\downarrow$ . It is time for our

version of a synthesis. In the proof-theoretic project, this means we back up until we find the sub-theses that led to our bad conclusion. We adjust those from  $\uparrow$  to  $\downarrow$  and we start over. We have a new thesis, or collection of theses, different from the original, and not simply a polarity reversal, but some new synthesis that we mark as  $\uparrow$  and proceed until we hit our next dead end and must reverse polarity again. Does this process ever end? No.

## **Derrida and *différance***

Derrida's concept of *différance*, with its deliberate orthographic gesture toward both difference and deferral, is in part an attempt to think negation and absence without the metaphysics of presence and absence that Heidegger and Sartre relied on. *Différance* is not itself a metaphysical thing. It is what makes presence and absence possible. The play of differences underlies all signification. The trace, the supplement, and the iterability of the sign are all part of an attempt to think the negative without grounding it in a metaphysics of negativity.

Derrida was correct to be suspicious of the idea that negation grounds itself in a metaphysical opposite, whether that be a real absence, a real Nothing, a real non-being. The problem is that he got rid of the metaphysics and retained the differential structure of classical negation. The genuine structure of negation is not differential but scalar. One travels up and down inferential scales. "Negation," such as it is, is the act of reasoning down those scales. Nothing in that idea shipwrecks Derrida's broader project. You can still be disruptive, but you now do it by introducing new scalar orderings.

The  $\downarrow$  account can execute Derrida's project, but more fruitfully. Negation is not differential; it marks an inferential direction relative to an ordering, and orderings are always contextual, always embedded in some inferential practice. They are never absolute. Negation still defers:  $\downarrow$  never bottoms out at some metaphysical ground of non-being, it propagates to a contextually determined depth and stops when we stop. The trace that Derrida finds in every sign (the presence of absence, the mark of what is not there) is just a way of saying that moving down a previous  $\downarrow$  inferential path brought us here. The traces are marks of varying depths of propagation (where we have been and where we are going).

## **The religious background**

It is worth asking why classical negation has had such a grip on both logicians and philosophers. The answer is partly technical; classical logic is mathematically clean, and the law of excluded middle is logically convenient, but it is also, as this essay argued in its opening sections, partly ideological. The picture of the world as divided into positive and negative facts, with every proposition having a determinate truth value and every predicate having a fully determinate complement, is not a discovery about the structure of reality. It is a choice – a choice that reflects, among other things, a picture of God as omniscient, as seeing the world from a standpoint at which every question has a determinate answer, including every question about what is not the case.

Brouwer saw this clearly. His rejection of the law of excluded middle was not merely a technical move in the foundations of mathematics – it was a rejection of a theological conception of

mathematics – the idea that mathematical truths exist independently of any process of construction and that every mathematical proposition is determinately true or false, whether or not we can verify it. The classical picture of negation is the theological picture: there is a God’s-eye view from which  $\neg P$  is fully determinate, from which the complement of every set is fully defined, from which every absence is as real as every presence.

The  $\downarrow$  account is resolutely non-theological. It does not assume a God’s-eye view. It does not assume that every question has a determinate answer. It does not assume that the complement of every predicate is fully defined.  $\downarrow$  is the perspective of a reasoner embedded in a context, reasoning from what they know toward what follows, marking the directions  $\uparrow$  and  $\downarrow$  in which inferences run and stopping at some contextually determined level of inferential depth. There is no determinate negation to serve as a metaphysical engine for theologians, or for philosophers such as Heidegger, Sartre, and Hegel.

What we have instead is simpler and more honest: an inference runs downward from here, as far as the context requires. That is all negation ever was.

## Coda

The great philosophical accounts of negation (Hegel’s dialectic, Heidegger’s Nothing, Sartre’s *négatités*, Derrida’s *différance*) are not entirely off course. They all begin with the idea that “negation” is a kind of human cognitive practice. The problem is that they wed this very sound idea to some version of classical negation and use that fusion to extrude an ungrounded philosophical thesis. As I said, the initial idea is not wrong (“negation” is in fact a cognitive practice), but this does not require classical  $\neg$ . It requires  $\downarrow$  – the capacity to run inferences downward through the orderings that structure our experience, to a depth determined by context and need and the limits of what we know. That is the human capacity. It may have genuine metaphysical consequences, but those consequences have yet to be explored.

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